Comsats University Islamabad

Attock Campus

Course Details

Course Name	***Information security***
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lab task #01

code:

```python
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

# Example
print(gcd(a: 48, b: 18))  # Output: 6
```

output:

```
C:\Users\Superuser\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Superuser\PycharmP
6

Process finished with exit code 0
```

lab task #02:

code:

```python
def gcd_with_steps(a, b):
    while b != 0:
        print(f"{a} = {b} * ({a // b}) + {a % b}")
        a, b = b, a % b
    print(f"GCD = {a}")
    return a

# Example
gcd_with_steps(a: 48, b: 18)
```

output:

```
C:\Users\Superuser\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Superuser\Pych
48 = 18 * (2) + 12
18 = 12 * (1) + 6
12 = 6 * (2) + 0
GCD = 6

Process finished with exit code 0
```

lab task 3:

```python
def gcd_recursive(a, b):  2 usages
    if b == 0:
        return a
    return gcd_recursive(b, a % b)

# Example
print(gcd_recursive(a: 48, b: 18))  # Output: 6
```